

Game Theory

Structure-only skeleton (course content pending)

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1 About these notes

1.1 Scope

These are lecture notes for *Game Theory* (third-semester elective, Master in Physics of Data, University of Padova). At the time of writing the professor has not yet distributed slides; this book is therefore a **structural skeleton** — chapter titles and a suggested organisation that will be filled in as the course progresses.

The skeleton follows the standard non-cooperative-then-cooperative curriculum:

- Chapter 1 — normal (strategic) form games; dominance and rationalisability.
- Chapter 2 — Nash equilibrium in pure strategies.
- Chapter 3 — mixed strategies and the existence theorem.
- Chapter 4 — extensive-form games and backward induction.
- Chapter 5 — games of incomplete information (Bayesian games).
- Chapter 6 — cooperative game theory; core, Shapley value.
- Chapter 7 — mechanism design and auction theory.

Every chapter is currently a stub with a suggested table of contents.

1.2 Sources (planned)

Primary references: Osborne & Rubinstein, *A Course in Game Theory*, MIT Press (1994); Fudenberg & Tirole, *Game Theory*, MIT Press (1991); Myerson, *Game Theory: Analysis of Conflict*, Harvard University Press (1991).

2 Strategic Form Games

Status: skeleton. Content will be added once the course delivers lecture material.

2.1 Planned outline

(to be filled in)

2.2 References

Osborne & Rubinstein; Fudenberg & Tirole; Myerson.

3 Nash Equilibrium

Status: skeleton. Content will be added once the course delivers lecture material.

3.1 Planned outline

(to be filled in)

3.2 References

Osborne & Rubinstein; Fudenberg & Tirole; Myerson.

4 Mixed Strategies And Existence

Status: skeleton. Content will be added once the course delivers lecture material.

4.1 Planned outline

(to be filled in)

4.2 References

Osborne & Rubinstein; Fudenberg & Tirole; Myerson.

5 Extensive Form Games

Status: skeleton. Content will be added once the course delivers lecture material.

5.1 Planned outline

(to be filled in)

5.2 References

Osborne & Rubinstein; Fudenberg & Tirole; Myerson.

6 Bayesian Games

Status: skeleton. Content will be added once the course delivers lecture material.

6.1 Planned outline

(to be filled in)

6.2 References

Osborne & Rubinstein; Fudenberg & Tirole; Myerson.

7 Cooperative Game Theory

Status: skeleton. Content will be added once the course delivers lecture material.

7.1 Planned outline

(to be filled in)

7.2 References

Osborne & Rubinstein; Fudenberg & Tirole; Myerson.

8 Mechanism Design

Status: skeleton. Content will be added once the course delivers lecture material.

8.1 Planned outline

(to be filled in)

8.2 References

Osborne & Rubinstein; Fudenberg & Tirole; Myerson.

